

PRACTICA 1 - ALGORITMICA

EJERCICIO 1

He creado el código añadiendo el resto de algoritmos mediante includes, y en base a los parametros de entrada se elige una función u otra.

```
/* Archivos con funciones */

#include "bubble.cpp"
#include "insertion.cpp"
#include "selection.cpp"
#include "mergesort.cpp"
#include "quicksort.cpp"
#include "heapsort.cpp"
#include "fibonacci.cpp"
#include "hanoi.cpp"

#include <iostream>
#include <cstdlib>
#include <ctime>
#include <fstream>
#include <string>
#include <cstring>

using namespace std;

int main(int argc, char *argv[]) {

    if (argc <= 3) {
        cerr << "Error: El programa se debe ejecutar de la siguiente forma:\n";
        cerr << argv[0] << " NombreAlgoritmo Semilla tamCaso1 tamCaso2 ...
tamCasoN\n";
        return 0;
    }

    int *vectorAlgoritmo;
    int tam, i;
    unsigned long int tiempoInicio, tiempoFin;
    double tiempoEjecucion;
    unsigned long int semilla;
    ofstream fsalida;
    string algoritmo;

    algoritmo = argv[1];

    fsalida.open("../dat/" + algoritmo + ".dat");
```

```
if (!fsalida.is_open()) {
    cerr << "Error: No se pudo abrir el fichero " << algoritmo << ".dat\n";
    return 0;
}

semilla = atoi(argv[2]);
srand(semilla);

for (int argumento = 3; argumento < argc; argumento++) {

    tam = atoi(argv[argumento]);

    vectorAlgoritmo = new int[tam];

    for (i = 0; i < tam; i++) {
        vectorAlgoritmo[i] = rand() % tam;
    }

    tiempoInicio = clock();

    if (algoritmo == "bubble")
        bubble(vectorAlgoritmo, tam);

    else if (algoritmo == "insertion")
        insertion(vectorAlgoritmo, tam);

    else if (algoritmo == "selection")
        seleccion(vectorAlgoritmo, tam);

    else if (algoritmo == "mergesort")
        mergesort(vectorAlgoritmo, tam);

    else if (algoritmo == "quicksort")
        quicksort(vectorAlgoritmo, tam);

    else if (algoritmo == "heapsort")
        heapsort(vectorAlgoritmo, tam);

    else if (algoritmo == "fibonacci")
        fibo(tam);

    else if (algoritmo == "hanoi")
        hanoi(tam, 1, 2);

    else {
        cerr << "Algoritmo no reconocido\n";
        return 0;
    }

    tiempoFin = clock();

    tiempoEjecucion = 1000.0 * (tiempoFin - tiempoInicio) / CLOCKS_PER_SEC;

    fsalida << tam << " " << tiempoEjecucion << "\n";
}
```

```
        cerr << "Tam: " << tam << " Tiempo: " << tiempoEjecucion << " ms" << endl;

        delete[] vectorAlgoritmo;

    }

    fsalida.close();
    return 0;

}
```

```
PS C:\Users\david\Desktop\Ingenieria Informatica 2\ALG\Practical\Codigo> ./main insertion 1234 1000 2500 5000 10000
Tam: 1000 Tiempo: 1 ms
Tam: 2500 Tiempo: 4 ms
Tam: 5000 Tiempo: 23 ms
Tam: 10000 Tiempo: 79 ms
```

EJERCICIO 2

En este ejercicio hemos ejecutado cada algoritmo con una serie de parametros y la semilla 1333. En la carpeta cap están todos los resultados, no pongo las imagenes ya que los resultados se usan en las tablas del Ejercicio 3.

EJERCICIO 3

- BUBBLE (n^2)

n	T(n) ms	f(n) operaciones	K(n) ms/operacion
1000	1	1000000	0.000001
2000	5	4000000	0.00000125
3000	9	9000000	0.000001
4000	15	16000000	0.0000009375
5000	25	25000000	0.000001
6000	34	36000000	0.0000009444
7000	46	49000000	0.0000009388
8000	58	64000000	0.00000090625
9000	77	81000000	0.0000009506
10000	99	100000000	0.00000099
11000	123	121000000	0.0000010165
12000	156	144000000	0.0000010833

n	T(n) ms	f(n) operaciones	K(n) ms/operacion
13000	185	169000000	0.0000010946
14000	226	196000000	0.0000011530
15000	268	225000000	0.0000011911
16000	336	256000000	0.0000013125
17000	378	289000000	0.0000013079
18000	431	324000000	0.0000013302
19000	475	361000000	0.0000013158
20000	518	400000000	0.000001295
22000	683	484000000	0.0000014111
24000	812	576000000	0.0000014097
26000	978	676000000	0.0000014467
28000	1125	784000000	0.000001435
30000	1325	900000000	0.0000014722

- **SELECTION (n^2)**

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
1000	1	1000000	0.000001
2000	2	4000000	0.0000005
3000	4	9000000	0.0000004444444444
4000	5	16000000	0.0000003125
5000	7	25000000	0.00000028
6000	11	36000000	0.0000003055555556
7000	14	49000000	0.0000002857142857
8000	19	64000000	0.000000296875
9000	25	81000000	0.0000003086419753
10000	29	100000000	0.00000029
11000	36	121000000	0.0000002975206612
12000	41	144000000	0.0000002847222222
13000	48	169000000	0.0000002840236686
14000	54	196000000	0.0000002755102041

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
15000	64	225000000	0.0000002844444444
16000	71	256000000	0.00000027734375
17000	82	289000000	0.0000002837370242
18000	92	324000000	0.0000002839506173
19000	103	361000000	0.0000002853185596
20000	114	400000000	0.000000285
22000	137	484000000	0.0000002830578512
24000	162	576000000	0.00000028125
26000	194	676000000	0.0000002869822485
28000	219	784000000	0.0000002793367347
30000	251	900000000	0.0000002788888889

- **INSERTION (n^2)**

n	T(n) ms	f(n) operaciones	K(n) ms/operacion
1000	0	1000000	0
2000	2	4000000	0.0000005
3000	5	9000000	0.0000005555555556
4000	8	16000000	0.0000005
5000	13	25000000	0.00000052
6000	17	36000000	0.0000004722222222
7000	23	49000000	0.0000004693877551
8000	31	64000000	0.000000484375
9000	38	81000000	0.0000004691358025
10000	48	100000000	0.00000048
11000	56	121000000	0.0000004628099174
12000	72	144000000	0.0000005
13000	78	169000000	0.0000004615384615
14000	92	196000000	0.0000004693877551
15000	103	225000000	0.0000004577777778
16000	122	256000000	0.0000004765625

n	T(n) ms	f(n) operaciones	K(n) ms/operacion
17000	134	289000000	0.0000004636678201
18000	149	324000000	0.0000004598765432
19000	171	361000000	0.0000004736842105
20000	189	400000000	0.0000004725
22000	233	484000000	0.0000004814049587
24000	262	576000000	0.0000004548611111
26000	325	676000000	0.0000004807692308
28000	349	784000000	0.0000004451530612
30000	402	900000000	0.0000004466666667

- **MERGESORT ($n * \log n$)**

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
5000	0	42585.96596	0
10000	1	92103.40372	0.00001085736205
15000	2	144237.0822	0.00001386605975
20000	3	198069.7511	0.00001514617949
25000	3	253165.7776	0.00001184994287
30000	3	309268.5798	0.000009700306451
35000	4	366208.6169	0.00001092273588
40000	5	423865.3893	0.00001179619786
45000	6	482148.7996	0.00001244429107
50000	7	540988.9142	0.00001293926699
60000	7	660125.9905	0.00001060403635
70000	9	780937.5365	0.00001152460931
80000	11	903182.5531	0.00001217915466
90000	12	1026680.845	0.00001168815027
100000	15	1151292.546	0.00001302883446
120000	15	1403429.643	0.00001068810259
140000	18	1658915.678	0.00001085046108
160000	20	1917268.655	0.00001043150627

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
180000	24	2178128.183	0.00001101863526
200000	27	2441214.529	0.00001106006853
220000	27	2706304.222	0.000009976705422
240000	33	2973214.609	0.00001109909789
260000	34	3241793.597	0.00001048802121
280000	37	3511912.567	0.00001053556981
300000	38	3783461.326	0.00001004371308

- **QUICKSORT ($n * \log n$)**

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
5000	0	42585.96596	0
10000	1	92103.40372	0.00001085736205
15000	1	144237.0822	0.000006933029875
20000	2	198069.7511	0.00001009745299
25000	2	253165.7776	0.00000789996191
30000	3	309268.5798	0.000009700306451
35000	3	366208.6169	0.000008192051911
40000	3	423865.3893	0.000007077718718
45000	3	482148.7996	0.000006222145534
50000	4	540988.9142	0.000007393866852
60000	5	660125.9905	0.00000757431168
70000	6	780937.5365	0.00000768307287
80000	7	903182.5531	0.000007750371147
90000	7	1026680.845	0.000006818087657
100000	8	1151292.546	0.00000694871171
120000	11	1403429.643	0.0000078379419
140000	11	1658915.678	0.000006630837326
160000	14	1917268.655	0.00000730205439
180000	15	2178128.183	0.000006886647037
200000	16	2441214.529	0.000006554114687

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
220000	17	2706304.222	0.00000628162934
240000	20	2973214.609	0.000006726725996
260000	22	3241793.597	0.000006786366665
280000	22	3511912.567	0.000006264392857
300000	23	3783461.326	0.000006079089494

- **HEAPSORT ($n * \log n$)**

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
5000	1	42585.96596	0.00002348191423
10000	1	92103.40372	0.00001085736205
15000	2	144237.0822	0.00001386605975
20000	2	198069.7511	0.00001009745299
25000	3	253165.7776	0.00001184994287
30000	4	309268.5798	0.00001293374194
35000	4	366208.6169	0.00001092273588
40000	5	423865.3893	0.00001179619786
45000	6	482148.7996	0.00001244429107
50000	6	540988.9142	0.00001109080028
60000	9	660125.9905	0.00001363376102
70000	10	780937.5365	0.00001280512145
80000	11	903182.5531	0.00001217915466
90000	12	1026680.845	0.00001168815027
100000	13	1151292.546	0.00001129165653
120000	18	1403429.643	0.00001282572311
140000	19	1658915.678	0.00001145326447
160000	20	1917268.655	0.00001043150627
180000	24	2178128.183	0.00001101863526
200000	26	2441214.529	0.00001065043637
220000	30	2706304.222	0.00001108522825
240000	34	2973214.609	0.00001143543419

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
260000	36	3241793.597	0.00001110496363
280000	42	3511912.567	0.00001195929545
300000	43	3783461.326	0.00001136525427

• **FIBONACCI (((1 + $\sqrt{5}$) / 2)^2)**

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
1	0	1.618033989	0
2	0	2.618033989	0
4	0	6.854101966	0
6	0	17.94427191	0
8	0	46.97871376	0
10	0	122.9918694	0
13	0	521.0019194	0
16	0	2206.999547	0
18	0	5777.999827	0
20	0	15126.99993	0
22	1	39602.99997	0.00002525061234
24	0	103682	0
25	0	167761	0
26	1	271443	0.000003684014692
28	1	710647	0.000001407168397
30	3	1860498	0.0000016124715
32	9	4870847	0.000001847727921
34	23	12752043	0.000001803632563
36	56	33385282	0.000001677385861
38	144	87403803	0.000001647525566
40	378	228826127	0.000001651909268
42	992	599074578	0.000001655887324
44	2597	1568397607	0.000001655830121
45	4222	2537720636	0.000001663697706

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
50	46985	28143753123	0.000001669464616

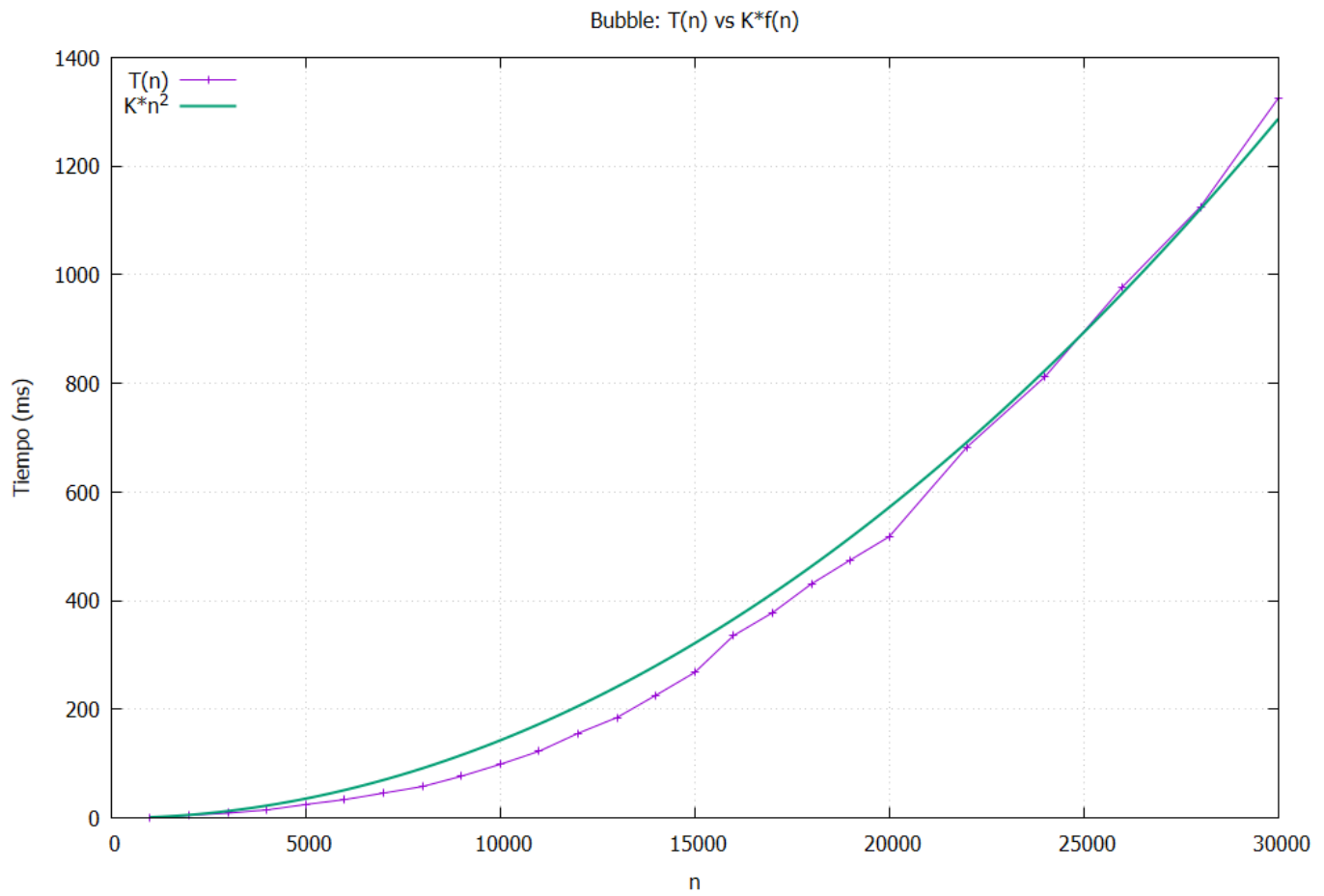
- **HANOI (2^n)**

n	T(n) ms	f(n) operaciones	K(n)=T(n)/f(n)
1	0	2	0
2	0	4	0
3	0	8	0
4	0	16	0
5	0	32	0
6	0	64	0
7	0	128	0
8	0	256	0
9	0	512	0
10	0	1024	0
11	0	2048	0
12	0	4096	0
13	0	8192	0
14	0	16384	0
15	0	32768	0
16	0	65536	0
17	1	131072	0.000007629394531
18	0	262144	0
19	1	524288	0.000001907348633
20	3	1048576	0.000002861022949
21	6	2097152	0.000002861022949
22	12	4194304	0.000002861022949
23	25	8388608	0.000002980232239
24	47	16777216	0.000002801418304
25	98	33554432	0.000002920627594

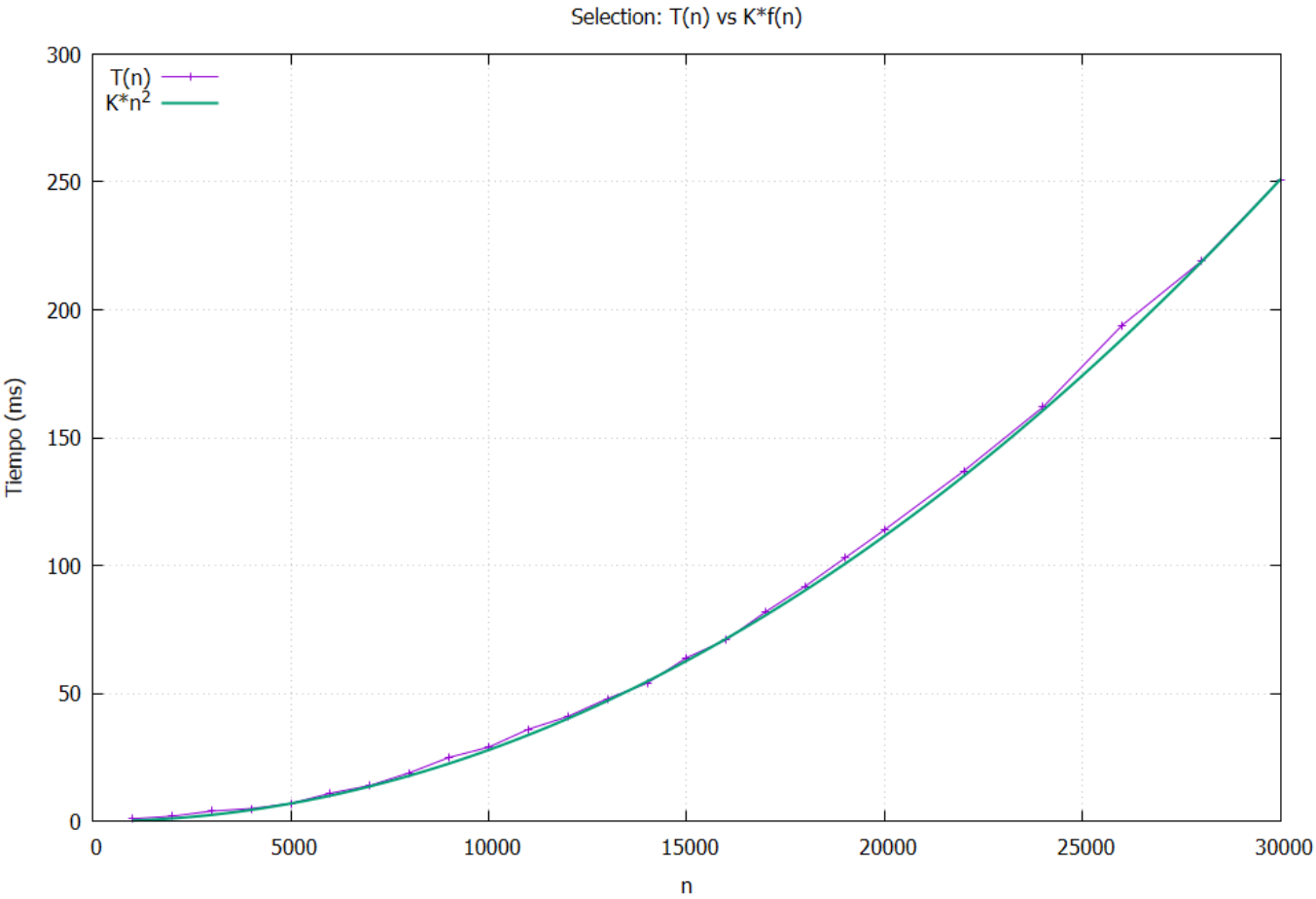
EJERCICIO 4

Finalmente mediante gnuplot hemos generado las gráficas de cada algoritmo con su tiempo real y su tiempo teórico.

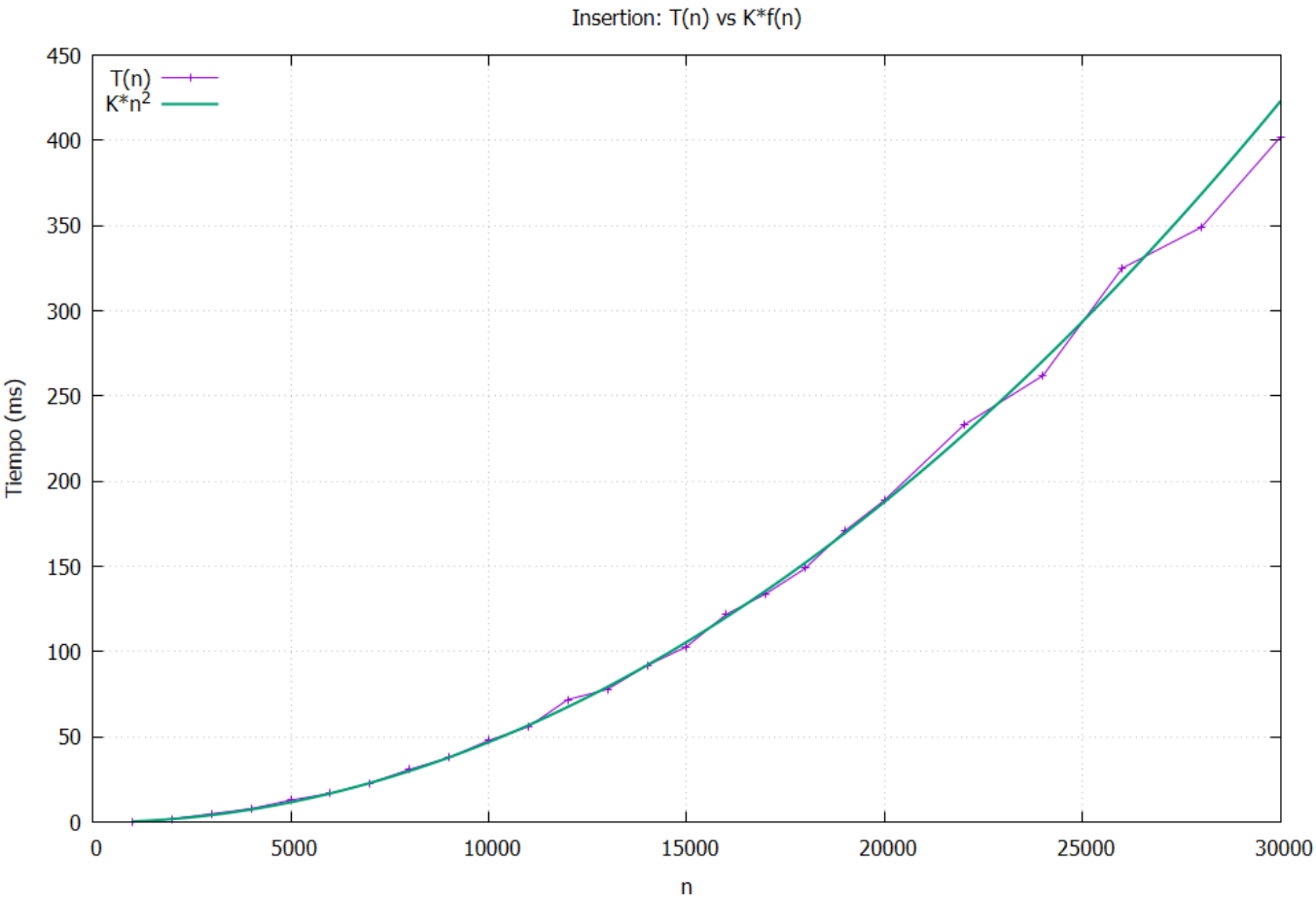
• **BUBBLE (n^2)**



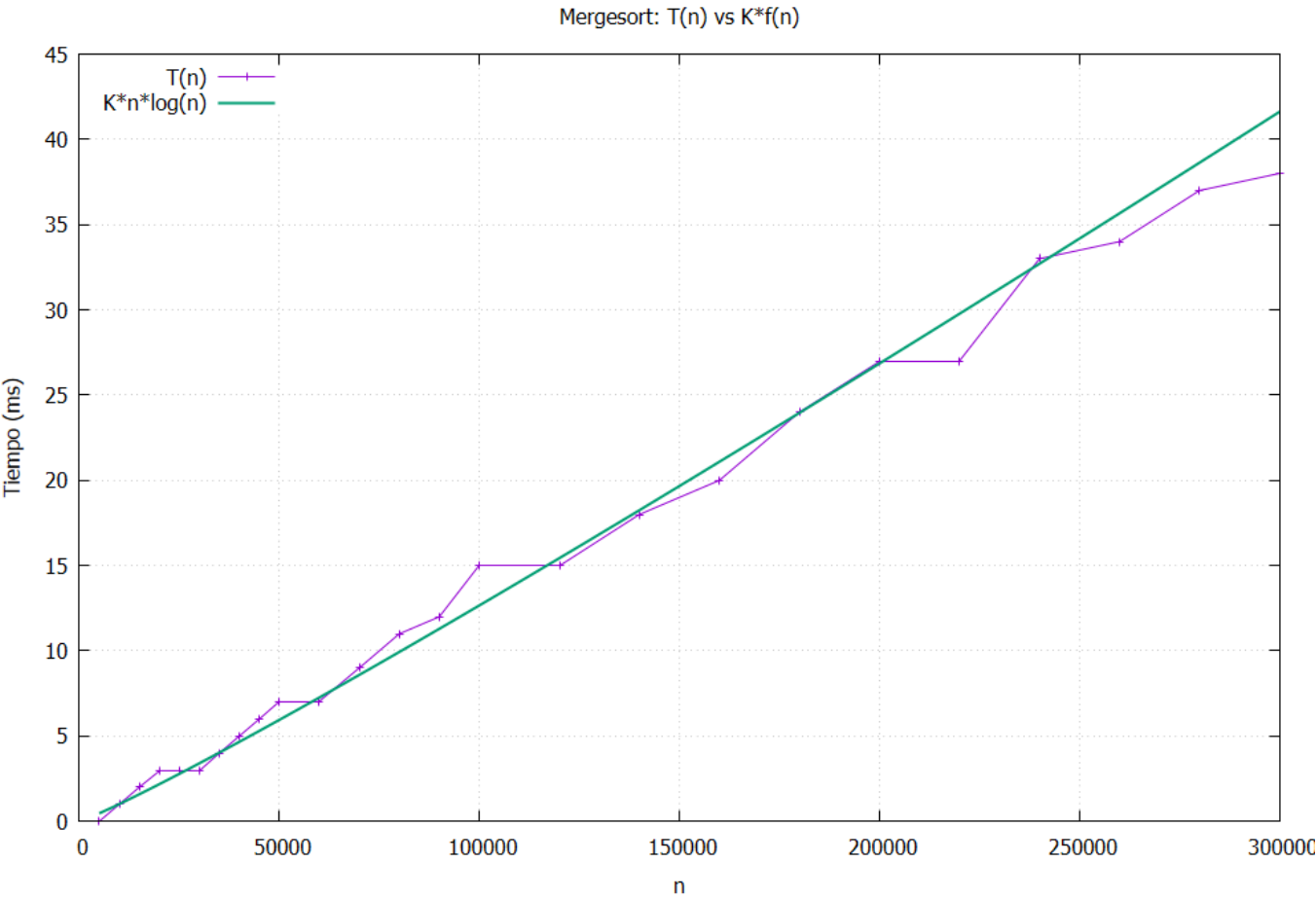
• **SELECTION (n^2)**



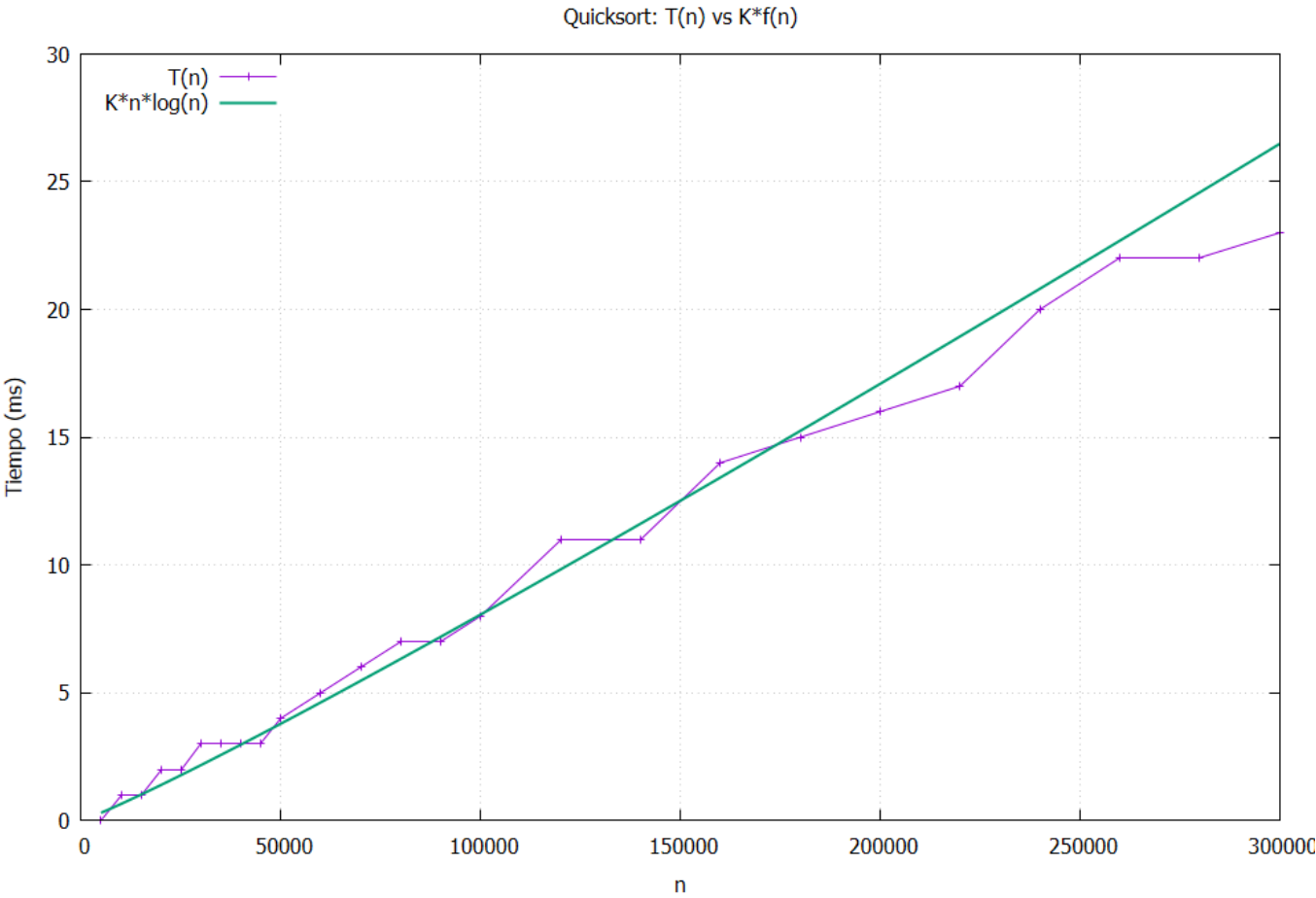
• INSERTION (n^2)



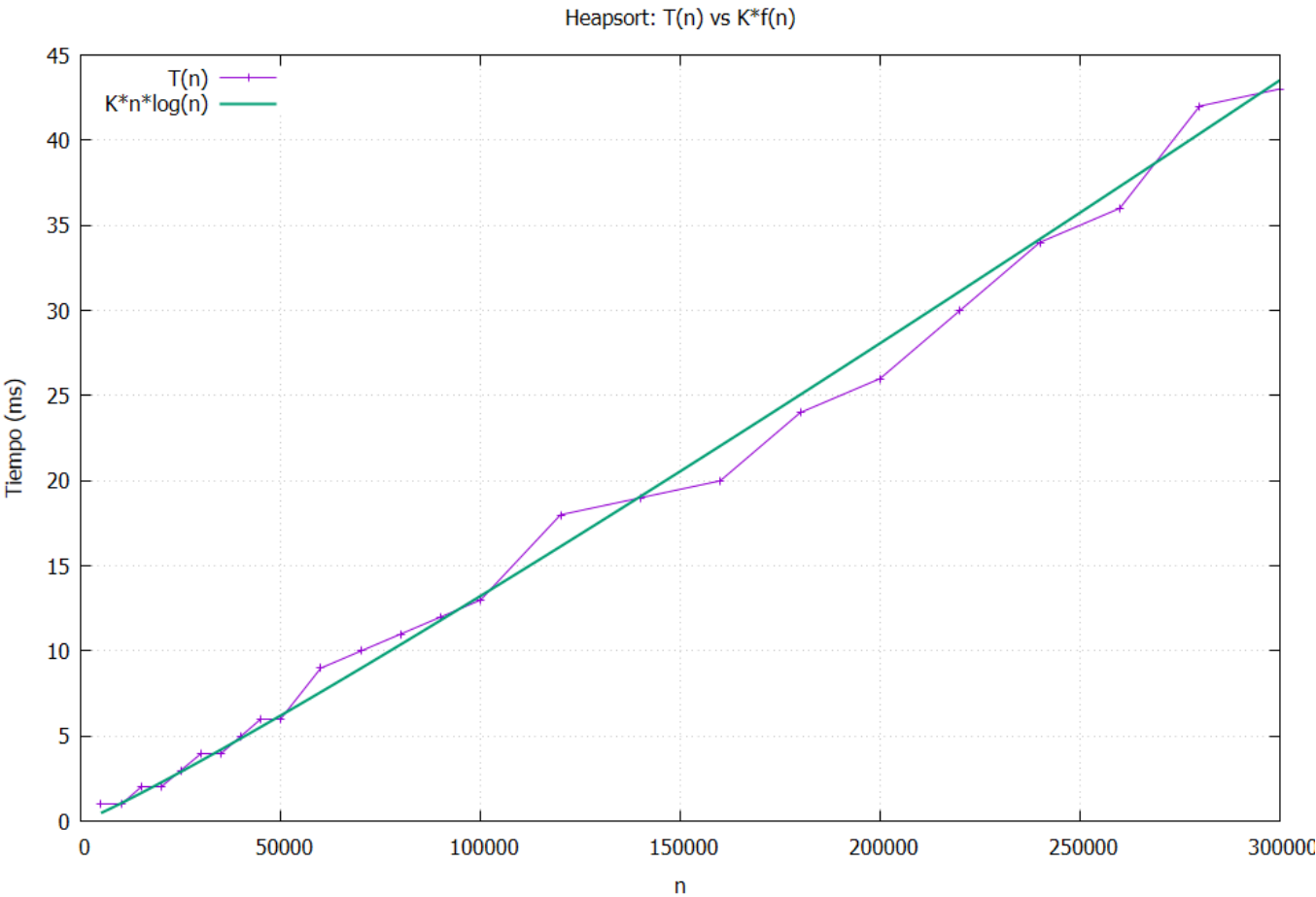
- MERGESORT ($n * \log n$)**



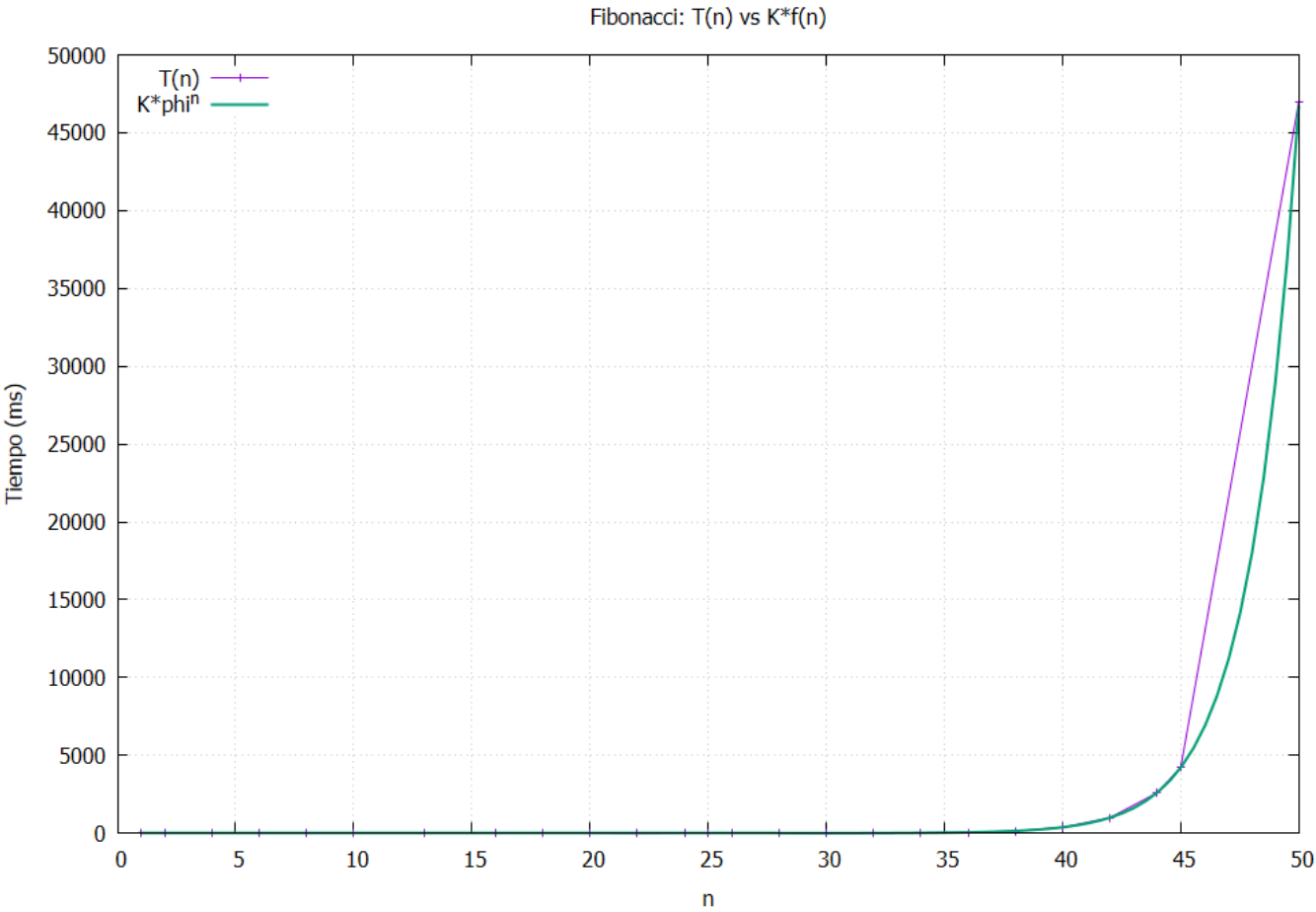
- QUICKSORT ($n * \log n$)**



- **HEAPSORT ($n * \log n$)**



• **FIBONACCI ($((1 + \sqrt{5}) / 2)^2$)**



• **HANOI (2^n)**

